

## ELECTIVE-5.1

### URA411: INDUSTRIAL SMART SENSOR

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**Course Objective:** This course would let the users to understand industrial sensors working principle and its applications in industrial process, covers the principles of operation of smart sensors, including IO-Link communication and how they can be used for the implementation of Industry4.0, explore the features of RFID system and learn how to write data on an RFID Tag for industrial applications.

#### Syllabus

**Sensors for Object Detection:** Introduction to Sensors, sensors in day to day life, Definition of the Term sensor, Milk bottling system project, Technical knowledge on sensors, Object detection in Industrial Applications, proximity switches – operating principle, proximity switch as binary sensor, proximity switches -applications, switching performance of proximity switches: switching distance, switching distance with inductive sensors, hysteresis, hysteresis with inductive sensor, preparation of problem: Lifting device, proximity switches connection technology, components of sensors, special designs of inductive sensor, Introduction of Optical sensors, Through beam sensors and its applications, Diffuse sensor, diffuse reflection sensor, Retro reflective sensor, Fibre optics, Red Light/ Infrared light, laser sensor, Types of reflection, Background suppression, Adjusting diffusive sensor, Datasheet entries, Contrast sensors, capacitive sensors and its design and mode of operation, Ultrasonic sensors: design and function, applications.

**Smart sensors for smart systems:** Introduction to Smart sensors, Basics of Smart sensors: Fundamentals of smart sensors, IIOT and smart sensors, Function and interfaces of smart sensors, advantages of smart sensors, pin allocation of smart sensor, communication terms of smart sensors.

Smart sensors in production: Design and functions of an ultrasonic Sensor, commissioning of smart sensors, parameterization of a smart laser Distance sensor, parameterization of smart sensors via the PLC, capacitive sensor, Environment and quality, smart sensors in a PLC program, IO-Link sensors, Data storage in smart sensors.

**Integration in Siemens PLC with an External IO-Link Master:** Introduction to siemens PLCs, Step 7 in TIA portal, setup and connections, Addition of PLC in the project, Addition of IO-Link Master to the project, project download to the PLC, configuration of ultrasonic sensor in the project, observing the process data of the sensor, reading acyclic data from the sensor, Writing acyclic data on the sensor, PLC integration of smart sensor.

**Integration in a siemens PLC with an IO-Link communication Module:** setup and connections of siemens PLC, Addition of PLC to the project, addition of the IO-link communication module to the project, project download to the PLC, configuration of the ultrasonic sensor in the project, observing the process data of the sensor, reading acyclic data from the sensor, writing acyclic data on the sensor, PLC integration of smart sensors.

**Pactware software:** Introduction to pactware, installing pactware, Importing IODD files, familiarization of pactware software and its features, study of smart sensors by connecting online with pactware software.

**Object identification with RFID:** Object identification, number systems, Introduction to RFID Technology, Basics of RFID Technology: RFID systems and their ranges, coupling methods, transmission frequencies, selection of RFID, RFID applications, object identification to product memory with RFID, RFID programming with PLC.

**HMI:** Basics of HMI: Benefits of the human senses, basic concepts for the ergonomic design of HMI, User-Machine interfaces, the process of developing a graphical user interface, the process of developing a graphical user interface, designing a graphical user interface, Objects for creating a graphical user interface in the TIA portal, HMI programming.

### **Laboratory Work**

1. Object detection in industrial applications: Switching distance, lifting device, Object material, conveyor belt,
2. Function and commissioning of Ultrasonic sensor, capacitive sensor and other smart sensors
3. Parametrization of smart laser distance sensor
4. Parametrization of smart sensors via PLC
5. IO-Link sensors commissioning and communication.
6. Understanding the working principle of smart inductive sensor
7. Data storage in smart sensors
8. Characteristics influence factors.
9. Writing data to an RFID Tag
10. Reading and analyzing the data on an RFID tag
11. Visualization of parameters in HMI

### **Course Learning Objectives (CLO)**

The students will be able to:

1. grasp the knowledge of basic mode of operation of sensors for object detection in production process and identify suitable sensor for an application.
2. learn how to add a PROFINET IO device with a GSDML file to the hardware catalog and connect it to a PLC to interface IO-Link sensors with PLC in a TIA portal and would be able to break down the process data of the sensor into the individual information.
3. implement data management mode and be able to use it for parameterization of the laser distance sensor.
4. demonstrate the interfaces of the function module IO\_LINK\_DEVICE for parameterization of a generic IO-Link device.
5. perform read or write access to acyclic data with the function module IO\_LINK\_DEVICE, reading or writing parameter data and can switch between several parameter sets for a smart sensor via HMI.

### **Text Books**

1. Object Identification with RFID, revision:03/2018. Authors: Theory section:Heike Greibl, Ramona sachse, Frank ebel, Dimitrios tsakas. Practical section: Dimitrios Tsakas, Ralph- Christoph Weber
2. Sensors, Actuators, and their Interfaces: A multidisciplinary introduction (Materials, Circuits and Devices), Nathan Ida, SciTech Publishing, 2011

### **Reference Books & Websites**

1. Fundamental principles of smart sensors: Author: Festo Didactic Staffs ,edition;11/2019
2. Industrial Smart sensors, LX
3. Smart sensor for smart systems, LX
4. Introduction to RFID Technology, LX
5. Basics of RFID, LX
6. HMI Operation and Visualization, LX
7. Plant Visualization with HMI, LX
8. LX.festo.com

### **Hardware Reference:**

1. Sensor for object detection kit